

A cut-cell immersed boundary method for static and moving bodies

in a collocated finite-volume solver

Formulation, implementation in OpenFOAM and verification

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Abstract

We present a cut-cell immersed boundary method for the incompressible Navier–Stokes equations on a fixed, non body-fitted mesh, its extension to rigid bodies moving through the mesh, and the transport of a passive scalar in the same framework. The body is described by a level set; each face of the mesh carries the fraction of its area that lies in the fluid and each cell the fraction of its volume. The wall segment of a cut cell is not computed independently but *defined* by the geometric closure of the fluid polyhedron, and this single identity carries the whole boundary treatment: the wall pressure force is absorbed into a face sum, no-penetration follows from discrete continuity, and the difference between a free-slip and a no-slip wall is one diagonal coefficient. No ghost values, no extrapolation into the solid and no coupling between velocity components appear. For a moving body only continuity changes: it acquires the flux swept by the wall as a source, and taking that flux — rather than the geometric volume change — as the primary quantity makes the discrete geometric conservation law hold by construction, so that a free stream is preserved to machine precision at any body speed. A passive scalar is transported by the same face fluxes and sees the wall either as a fixed-value or as a zero-flux surface, and its budget closes to round-off. The method is implemented in OpenFOAM without modifying the mesh: the fractions enter the standard operators explicitly, which keeps the pressure–velocity coupling of `simpleFoam`/`pimpleFoam` and the linear algebra untouched, and the whole moving-body extension sits behind a single run-time switch. The implementation is verified against an independent reference implementation of the same discretisation on a cylinder at $Re_D = 1$ at rest, oscillating and translating through a periodic seam, and by conservation tests for the scalar.

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1 Introduction

Immersed boundary methods let a body of arbitrary shape be represented on a mesh that does not conform to it. Among them, cut-cell methods are distinguished by being conservative: the fluid part of every cell is a well-defined control volume, and fluxes are accounted for on its actual boundary. Their classical difficulties are the description of the cut geometry, the small-cell problem, and, for moving bodies, the consistency between the change of the fluid volume of a cell and the flux swept by its wall, i.e. the discrete geometric conservation law.

The method described here addresses these points with a minimum of machinery. Its central object is the *closure identity* of a cut cell: the wet parts of its faces and its wall segment form a closed surface, so the wall area vector equals minus the sum of the wet face area vectors. Rather than computing the wall segment from the geometry and then enforcing consistency, the method defines the wall segment through the identity. Every consequence of the boundary condition — pressure force, no-penetration, wall traction, and for a moving body the volume swept — is then expressed with the same face fractions that the flux operators use, and the resulting discrete properties (exactness for uniform flow, no spurious force from a uniform pressure, exact compatibility of the pressure system, free-stream preservation) hold to round-off by construction.

Section 3 defines the geometry and its computation, Section 4 the discretisation for a body at rest, Section 5 the extension to moving bodies, Section 6 the transport of a passive scalar. Section 7 describes the implementation in OpenFOAM, Section 8 the verification, and Section 10 the limitations.

2 Governing equations

We solve the incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p + \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

with p the kinematic pressure and $\mathbf{g}$ a body force, in a domain Ω containing one or more rigid bodies B_i with prescribed velocity $\mathbf{u}_b(\mathbf{x}, t)$ (zero for a body at rest, $\mathbf{U}(t) + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c}(t))$ for a rigid body). On each body surface one of two conditions is imposed: *no slip*, $\mathbf{u} = \mathbf{u}_b$, or *free slip*, $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n} = 0$ together with zero tangential traction. A passive scalar T obeys

$$\frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{u}T) - \nabla \cdot (D\nabla T) = q, \quad (2)$$

with, on the body, either a fixed value $T = T_w$ or zero diffusive flux $\partial T / \partial n = 0$.

The discretisation is a collocated finite-volume method with SIMPLE(C) pressure-velocity coupling in the arrangement of OpenFOAM's `simpleFoam` (steady) and `pimpleFoam` (transient, implicit Euler with outer iterations). We write V_P for the volume of cell P , $\mathbf{S}_f$ for the area vector of face f pointing out of P , ϕ_f for the volume flux through f , and $\delta_f = 1/|\mathbf{d}_f|$ for the inverse of the centre-to-centre distance across f . Face values are linear interpolates, $(\cdot)_f = \frac{1}{2}[(\cdot)_P + (\cdot)_N]$ on a uniform mesh.

3 Geometric description of the body

3.1 Fluid fractions and the closure identity

Definition 1 (Level set and fractions). *A body is given by a function $\varphi(\mathbf{x})$, negative inside the body and positive in the fluid; only its sign and its zero level are used. For a face f and a cell P ,*

$$\theta_f = \frac{|\{\mathbf{x} \in f : \varphi > 0\}|}{|f|} \in [0, 1], \quad \alpha_P = \frac{|\{\mathbf{x} \in P : \varphi > 0\}|}{V_P} \in [0, 1] \quad (3)$$

are the fluid area and volume fractions. A cell is fluid if $\alpha_P = 1$, solid if $\alpha_P = 0$, cut otherwise. Several bodies are combined through the minimum of their level sets, which represents their union exactly.

The fluid part Ω_P of a cut cell is a polyhedron bounded by the wet parts of the cell faces, of area vectors $\theta_f \mathbf{S}_f$, and by the wall segment, of area A_{wall} and outward unit normal $\mathbf{n}_w$ (Fig. 1).

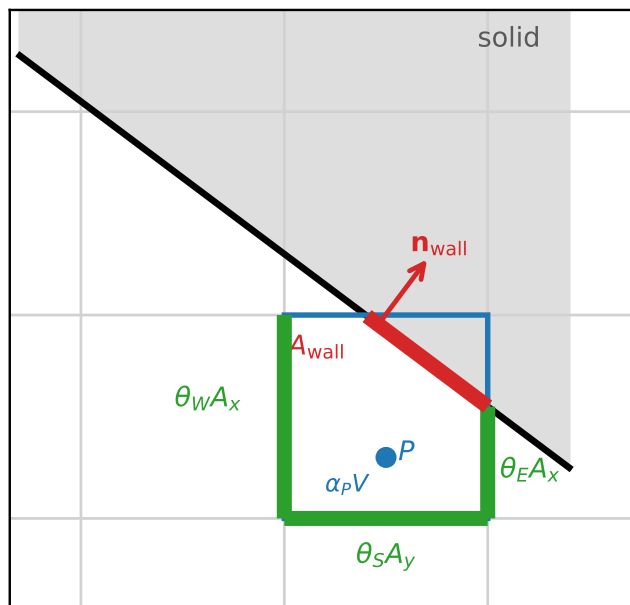


Figure 1: A cut cell. The fluid boundary consists of the wet parts of the faces (areas $\theta_f|\mathbf{S}_f|$) and the wall segment (area A_{wall} , outward normal $\mathbf{n}_w$). The cell carries fluid volume $\alpha_P V_P$.

Proposition 1 (Closure). *For every cut cell, $\sum_f \theta_f \mathbf{S}_f + \mathbf{n}_w A_{\text{wall}} = \mathbf{0}$.*

Proof. Apply the divergence theorem to a constant vector field over Ω_P . $\square$

We write $\mathbf{S}_w \equiv -\sum_f \theta_f \mathbf{S}_f = \mathbf{n}_w A_{\text{wall}}$ for the wall area vector, pointing from the fluid into the body, and $A_{\text{wall}} = |\mathbf{S}_w|$. In the implementation $\mathbf{S}_w$ is never computed from the level set: it is *defined* by Proposition 1 from the same θ_f that weight every flux operator, so the identity holds to round-off, however the wall is inclined and, for a moving body, however fast it moves.

Two further derived quantities are needed. The effective wall-normal distance of the cell average from the wall is estimated from the volume the cell holds behind its wall segment, halved for a profile linear in the wall-normal direction,

$$d_{\text{wall}} = \frac{\alpha_P V_P}{2A_{\text{wall}}}, \quad (4)$$

and the *Dirichlet wall coefficient* $A_{\text{wall}}/(d_{\text{wall}} V_P) = 2A_{\text{wall}}^2/(\alpha_P V_P^2)$, which multiplies a diffusivity to give the wall exchange coefficient per unit volume. A fluid cell one of whose faces is fully closed by a solid neighbour has $\mathbf{S}_w \neq \mathbf{0}$ as well; it is treated as a wall cell with $d_{\text{wall}} = h/2$, the staircase limit of the same formula.

3.2 Computation

The fractions are computed once for a body at rest and at every time step for a moving body, as follows.

1. The level set is evaluated at the mesh points, the face centres and the cell centres.
2. On every edge whose end points differ in sign, the crossing is located by bisection of the exact level set (40 steps, i.e. to machine precision relative to the edge length). The bisection runs in a canonical direction along the edge so that the two sides of a processor or periodic boundary obtain the same crossing.
3. θ_f is the area of the fluid polygon of the face (fluid vertices and crossing points, in order) divided by $|\mathbf{S}_f|$.
4. α_P is obtained by sub-sampling the level set on a lattice of n_s^d points in the cell ($n_s = 32$ here; d is the number of solution directions, so a two-dimensional case samples a 32×32 lattice), for every cell whose vertices, face centres and centre do not all lie on the same side. A body smaller than a cell is not resolved.
5. *Sliver absorption.* Cells with $\alpha_P < \alpha_{\min}$ (here 0.02) are absorbed into the body: $\alpha_P := 0$ and *all their faces are closed*, $\theta_f := 0$. Closing the faces is essential: an absorbed cell with an open face would break Proposition 1 for its neighbours, which would then see a flux into a cell that carries no equation. The absorption displaces the boundary by $O(\alpha_{\min} h)$, within the first-order error of the no-slip treatment.
6. *Coupled faces.* The two copies of a face on a periodic or processor boundary must carry the same fraction. After absorption, θ_f is synchronised across coupled faces with the minimum, which also propagates the closing of absorbed cells across the boundary. Doing this *after* absorption matters (Section 8.2).

Bodies described by a closed triangulated surface use the signed distance to the surface as level set (nearest point and inside/outside test), so that the same bisection applies. A body given only as a fluid-fraction field α has its face fractions reconstructed from the 0.5 iso-surface of the point-interpolated field; this route is less accurate (Section 10) and is intended for fractions produced by another model.

3.3 Multiple bodies

The level sets of all bodies are combined by their minimum before the fractions are computed, so overlapping bodies form an exact union. Every cut, solid or wall cell is attributed to a body — the one with the smallest level set at its centre, or the one across its closed face — which selects the wall condition (no slip or free slip; fixed value or zero flux for the scalar), the wall velocity, and the body to which its force contribution is added.

4 Discretisation for a body at rest

Every term of (1) is integrated over the fluid part Ω_P of each cell. The wall segment contributes to each term according to the boundary condition, and this is the only place where the boundary condition enters. Table 1 collects the result.

Table 1: Volume-integrated terms on the fluid part of a cell; Γ is a diffusivity, p_f and $\bar{\mathbf{u}}_f$ are linear face interpolates.

term	wall contribution	discrete operator
unsteady	—	$\alpha_P V_P (\mathbf{u}_P^{n+1} - \mathbf{u}_P^n) / \Delta t$
convective	zero ($\mathbf{u} \cdot \mathbf{n}_w = 0$)	$\sum_f \phi_f \bar{\mathbf{u}}_f, \quad \phi_f = \theta_f \bar{\mathbf{u}}_f \cdot \mathbf{S}_f$
viscous, free slip	zero (no tangential traction)	$\sum_f \theta_f \Gamma_f \delta_f \mathbf{S}_f (\mathbf{u}_N - \mathbf{u}_P)$
viscous, no slip	$\nu (\mathbf{0} - \mathbf{u}_P) A_{\text{wall}} / d_{\text{wall}}$	same, plus $-\nu A_{\text{wall}} / d_{\text{wall}} \mathbf{u}_P$
pressure	$p_P \mathbf{S}_w$, absorbed by closure	$\sum_f \theta_f (p_f - p_P) \mathbf{S}_f$
continuity	zero	$\sum_f \phi_f = 0$

The pressure entry deserves a derivation. The volume-integrated gradient is $\oint_{\partial\Omega_P} p \mathbf{n} dS$; splitting the boundary into the wet faces and the wall segment, and taking $p_w = p_P$ on the wall to first order,

$$\oint p \mathbf{n} dS = \sum_f \theta_f p_f \mathbf{S}_f + p_P \mathbf{S}_w \stackrel{\text{Prop. 1}}{=} \sum_f \theta_f (p_f - p_P) \mathbf{S}_f. \quad (5)$$

The wall pressure force — the form drag — is therefore absorbed exactly into the face sum, and a uniform pressure produces no force on any cell.

4.1 Properties

Proposition 2 (Exactness for uniform tangential flow). *For a plane wall and a uniform discrete velocity $\mathbf{u}$ with $\mathbf{u} \cdot \mathbf{n}_w = 0$, the discrete divergence vanishes identically in every cut cell.*

Proof. $\sum_f \phi_f = \mathbf{u} \cdot \sum_f \theta_f \mathbf{S}_f = -\mathbf{u} \cdot \mathbf{S}_w = 0.$ □

Proposition 3 (No-penetration is implied, not imposed). *For a locally uniform $\mathbf{u}$ in a cut cell with $A_{\text{wall}} \neq 0$, discrete continuity $\sum_f \phi_f = 0$ holds if and only if $\mathbf{u} \cdot \mathbf{n}_w = 0$.*

Proposition 3 is the structural point of the method: the no-penetration condition is never written down, it is a consequence of enforcing mass conservation on the cut cell with fluxes and wall segment taken from the same fractions. The free-slip condition appears only as the *absence* of a wall term in the viscous operator, and the no-slip condition as the presence of the single diagonal term $\nu A_{\text{wall}} / d_{\text{wall}}$. Because $d_{\text{wall}} \propto \alpha_P$, this coefficient grows like $1/\alpha_P$ as a cell is squeezed against the wall: the small-cell limit *strengthens* diagonal dominance and drives $\mathbf{u}_P$ onto the wall value rather than degrading the system. Consequently the discretisation contains no ghost value, no extrapolation into the solid, no blanking of cut cells, no coupling between the velocity components, and no adjustable parameter other than α_{min} .

Measured against exact solutions of Poisson problems on a disk, the free-slip treatment is second-order accurate and the no-slip treatment first-order accurate, the difference being entirely due to the wall-normal distance estimate (4), which free slip does not need.

4.2 Solid cells and the pressure–velocity coupling

Every face of a solid cell is closed, so its rows in the momentum and pressure matrices are empty. The momentum row receives a blanking diagonal $\nu \sum_f |\mathbf{S}_f| \delta_f / V_P$ (the viscous diagonal of an unobstructed cell) and the source that drives the cell to the body velocity; the pressure row receives a diagonal of the typical magnitude of the fluid rows and a zero source, so that $p = 0$ there. Neither value affects the fluid: the rows are decoupled.

The coupling is SIMPLEC. Writing the assembled momentum equation of cell P as $a_P \mathbf{u}_P = \mathbf{H}(\mathbf{u}) - \nabla_V p$, with $\nabla_V p$ the volume-integrated gradient (5) and $\mathbf{H} = \mathbf{b} - \sum_{nb} a_{nb} \mathbf{u}_{nb}$,

$$r_A = \frac{V_P}{a_P}, \quad r_{At} = \frac{V_P}{a_P + \sum_{nb} a_{nb}}, \quad \sum_f \theta_f (r_{At})_f \delta_f |\mathbf{S}_f| (p_N - p_P) = \sum_f \phi_{H,f} + S_P, \quad (6)$$

where $\phi_{H,f} = \theta_f (\mathbf{H}/a_P)_f \cdot \mathbf{S}_f + \theta_f (r_{At} - r_A)_f \delta_f |\mathbf{S}_f| (p_N^{\text{old}} - p_P^{\text{old}})$ is the Rhie–Chow flux of the predictor, and S_P is the mass source of a moving wall (zero for a body at rest, Section 5). The corrected flux $\phi_f = \phi_{H,f} - \theta_f (r_{At})_f \delta_f |\mathbf{S}_f| (p_N - p_P)$ is then discretely divergence-free, and the velocity is corrected with the gradient (5). Laplacian and flux in (6) are weighted by the same θ_f , so no inconsistency appears at the wall. The denominator of r_{At} is floored at the transient diagonal $\alpha_P V_P / \Delta t$ (the value of an unobstructed cell) to keep the pressure Laplacian definite.

The fluid region is a pure Neumann island for the pressure, and the reference is imposed as in `fvMatrix::setReference`, by doubling one diagonal entry rather than replacing a row. With L the pressure matrix on the fluid cells ($\mathbf{1}^\top L = \mathbf{0}$) and $L' = L + d_r \mathbf{e}_r \mathbf{e}_r^\top$, left-multiplying $L' p = \mathbf{b}$ by $\mathbf{1}^\top$ gives $d_r p_r = \mathbf{1}^\top \mathbf{b}$: for a compatible right-hand side the reference is transparent and $L p = \mathbf{b}$ exactly, whereas any incompatibility is deposited at the reference cell as a spurious point source. Exact compatibility of S_P is therefore not a nicety but a requirement (Section 5.2).

4.3 Force on the body

By closure, the pressure force on a body is $\mathbf{F}_p = \sum_P p_P \mathbf{S}_w$ over its wall cells, and the viscous force is the reaction to the traction already assembled in the momentum equation, $\mathbf{F}_v = \sum_P \nu A_{\text{wall}} / d_{\text{wall}} (\mathbf{u}_P - \mathbf{u}_b)$ for no slip and zero for free slip. Summing the momentum equations over the fluid shows that, at steady state in a periodic domain, the total force equals the driving body force times the fluid volume exactly; this identity is used as a consistency check in Section 8.

5 Moving bodies

5.1 What the wall sweeps

The mesh never moves: no face carries a mesh velocity and no mesh-motion term appears in any operator, so the method is not an arbitrary Lagrangian–Eulerian scheme. Only θ_f , α_P and $\mathbf{S}_w$ depend on time, and they are recomputed from the level set at every time step. The entire modification lives on the wall segment.

Let $\Omega_P(t)$ be the fluid part of cell P . Its boundary consists of the wet parts of the faces, which lie in fixed planes and sweep no volume, and the wall segment, which moves with $\mathbf{u}_b$. Reynolds transport gives

$$\frac{d(\alpha_P V_P)}{dt} = \int_{\text{wall}} \mathbf{u}_b \cdot \mathbf{n}_w dS \simeq \mathbf{u}_b \cdot \mathbf{S}_w, \quad (7)$$

and continuity over the same region, with $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n}_w = 0$ on the wall,

$$\boxed{\sum_f \theta_f \phi_f = -\mathbf{u}_b \cdot \mathbf{S}_w \equiv -S_P.} \quad (8)$$

The sign is as it must be: a body advancing into the fluid has $\mathbf{u}_b \cdot \mathbf{n}_w < 0$, the cell loses volume, and the net outflow through its faces is positive.

Table 2: Every term for a body at rest and a body moving. Only three rows differ.

term	body at rest	body moving
continuity	$\sum_f \theta_f \phi_f = 0$	$\sum_f \theta_f \phi_f = -S_P, \quad S_P = \mathbf{u}_b \cdot \mathbf{S}_w$
convective	no wall term	no wall term
pressure	$p_P \mathbf{S}_w$, absorbed by closure	same
viscous, free slip	no wall term	no wall term
viscous, no slip	$\nu(\mathbf{0} - \mathbf{u}_P) A_{\text{wall}}/d_{\text{wall}}$	$\nu(\mathbf{u}_b - \mathbf{u}_P) A_{\text{wall}}/d_{\text{wall}}$
unsteady	$\alpha_P V_P (\mathbf{u}^{n+1} - \mathbf{u}^n)/\Delta t$	$(\alpha_P^{n+1} \mathbf{u}^{n+1} - \alpha_P^n \mathbf{u}^n) V_P/\Delta t$

Table 2 lists every term. The convective row is the one that is easy to get wrong: the absolute mass flux through the wall is not zero, but the momentum balance over the *time-varying* region involves the relative flux $\mathbf{u}(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n}$, with $\mathbf{w}$ the boundary velocity, and on the wall $\mathbf{w} = \mathbf{u}_b$ and $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n} = 0$; on the fixed faces $\mathbf{w} = \mathbf{0}$ and the flux is the absolute one. So the convective operator is untouched, provided the unsteady term is the two-level one of Table 2 with the volume inside the difference. For free slip nothing else is added: repeating the argument of Proposition 3 with (8) forces $\mathbf{u} \cdot \mathbf{n}_w = \mathbf{u}_b \cdot \mathbf{n}_w$ for a locally uniform field — *the mass source is the boundary condition*. For no slip the wall traction $\nu(\mathbf{u}_b - \mathbf{u}_P) A_{\text{wall}}/d_{\text{wall}}$ keeps its diagonal half unchanged and acquires the source $\nu \mathbf{u}_b A_{\text{wall}}/d_{\text{wall}}$, which vanishes for a wall at rest.

5.2 Which quantity is primary

Equations (7) and (8) are the same statement, so on paper either may be taken as the definition. Discretely the two choices are not equivalent, and only one of them works.

Geometry first. Compute α^n and α^{n+1} from the level set and set $S_P = (\alpha_P^{n+1} - \alpha_P^n) V_P/\Delta t$. This differentiates a sampled quantity: the sub-sampled α carries a quadrature error of order 10^{-3} per cell, which divided by Δt is amplified without bound, and the resulting source is not compatible ($\sum_P S_P \neq 0$), so the whole defect is deposited at the pressure reference cell. The measured free-stream error of this variant is 50% of the body velocity and does not decrease with the number of outer iterations: it is a wrong fixed point, not a splitting error.

Flux first. Compute $\mathbf{S}_w$ from θ^{n+1} , set

$$S_P = \mathbf{u}_b \cdot \mathbf{S}_w, \quad \text{then} \quad \alpha_P^n V_P := \alpha_P^{n+1} V_P - \Delta t S_P. \quad (9)$$

Nothing is differentiated. $\sum_P S_P$ telescopes to round-off for every body position and every pattern of absorbed slivers (each interior face appears twice with opposite normals and the same θ_f , coupled faces are synchronised, and an absorbed cell has $\mathbf{S}_w = \mathbf{0}$), and for a uniform $\mathbf{u} \equiv \mathbf{u}_b$ the discrete divergence equals $-S_P$ *identically*, because $\mathbf{S}_w$ is assembled from the same θ_f that the flux uses. The second half of (9) is a backward Euler integration of (7): the classical resolution of the geometric conservation law, *take the volume from the flux, not the flux from the volume*, applied per step. Since α^n is recomputed from the current geometry at every step and never accumulated, it cannot drift; it converges to the geometric fraction at the old time at first order.

Proposition 4 (Free-stream preservation). *Let the body translate rigidly at $\mathbf{u}_b$ and let $\mathbf{u}^n \equiv \mathbf{u}_b$, $p \equiv 0$. Then $\mathbf{u}^{n+1} \equiv \mathbf{u}_b$, $p \equiv 0$ solves the discrete system exactly, for any Δt , body path, mesh and wall condition.*

Proof. In the momentum row of a fluid cell the row sums are: unsteady $\alpha^{n+1}V_P/\Delta t$; convective $\sum_f \phi_f = -S_P$ (the flux having been formed from $\mathbf{u}_b$ through the new θ_f , Section 5.3); viscous zero, since the θ -weighted Laplacian annihilates a constant; no-slip wall $+\nu A_{\text{wall}}/d_{\text{wall}}$ on the diagonal, matched by $+\nu A_{\text{wall}}\mathbf{u}_b/d_{\text{wall}}$ on the right; pressure zero by (5). The row reduces to $(\alpha^{n+1}V_P/\Delta t - S_P)\mathbf{u}_b = \alpha^n V_P \mathbf{u}_b/\Delta t$, which is (9). Continuity holds because $\sum_f \phi_f = -S_P$ identically, so the pressure equation has zero right-hand side and $p \equiv 0$. $\square$

Remark 1. The diagonal contribution of the convective operator is $\frac{1}{2} \sum_f \phi_f = -\frac{1}{2} S_P$, so unsteady and convective terms together put $\frac{1}{2}(\alpha^{n+1} + \alpha^n)V_P/\Delta t$ on the diagonal, the *average* of old and new fluid volume. A cell just emerging from the body still carries half its new volume there.

5.3 Algorithm

One time step (Fig. 2):

1. Move the bodies to $t + \Delta t$ and rebuild θ_f , α_P , $\mathbf{S}_w$, A_{wall} , d_{wall} , the cell sets and the body velocity at the cell centres $\mathbf{u}_b(\mathbf{x}_P)$.
2. Form $S_P = \mathbf{u}_b \cdot \mathbf{S}_w$ and $\alpha_P^n V_P = \alpha_P^{n+1} V_P - \Delta t S_P$.
3. *Re-interpolate the carried flux:* $\phi_f := \theta_f^{n+1} \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f$. The flux returned by the previous step lives on the faces of the *old* cut mesh: a face that has just opened has no history there, and one that has just closed carries a history that no longer exists. The re-interpolated flux is only the initial guess of the convective operator — the flux the step returns is the Rhie–Chow projected one either way — but for a fluid translating rigidly with the body it lands exactly on the fixed point, which is why Proposition 4 holds from the first outer iteration. Without it the fixed point is approached at roughly a factor of two per outer iteration.
4. For each of n_{outer} outer iterations: assemble and solve the momentum predictor with the sources $\alpha^n V_P \mathbf{u}^n/\Delta t$, $\nu A_{\text{wall}} \mathbf{u}_b/d_{\text{wall}}$ (no slip) and the blanking source; form $\mathbf{H}/a_P$ and ϕ_H ; solve (6) with S_P on the right-hand side; correct flux and velocity.

Equation (6) with S_P is the entire algorithmic modification. The pressure reference cell is kept while it is fluid and otherwise replaced by the fluid cell farthest from the bodies, so that a body cannot swallow it.

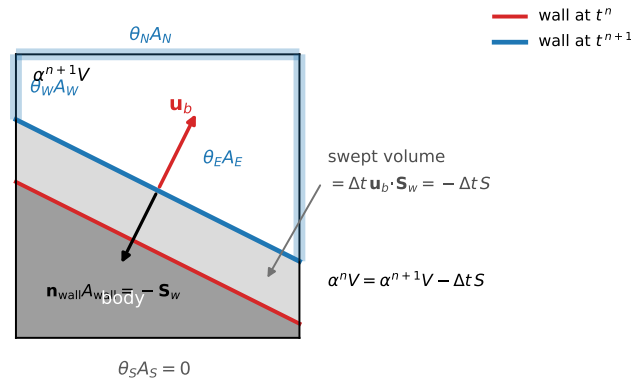


Figure 2: A cut cell over one step. The faces are fixed in space; only their wet fractions change. The shaded band is the volume the wall sweeps, $\Delta t \mathbf{u}_b \cdot \mathbf{S}_w = \Delta t S_P$, the only quantity the motion contributes to the discretisation.

5.4 Step restriction

The row sum of unsteady plus convective operator is exactly $\alpha^n V_P / \Delta t$, which changes sign when the swept volume exceeds the fluid the cell holds:

$$\Delta t |\mathbf{u}_b| \lesssim \frac{\alpha_P V_P}{A_{\text{wall}}} = 2d_{\text{wall}}. \quad (10)$$

This is a condition on the smallest wet cell, hence set by $\alpha_{\min}$ rather than by the mesh spacing. It is not fatal when violated in a few cells — the diagonal there is still dominated by the viscous and wall terms, which grow like $1/\alpha_P$ — and the solver reports $\min_P \alpha_P^n$ and the number of violating cells at every step.

5.5 Rigid body kinematics

A body is defined at rest and moved by a rigid motion: translation $\mathbf{d}(t)$ with velocity $\mathbf{U}(t) = \dot{\mathbf{d}}$, optionally a rotation with constant angular velocity $\boldsymbol{\omega}$ about its centre. The velocity must be the exact derivative of the displacement: the first supplies the wall condition, the second the mass source, and if they disagree the two halves of the boundary condition pull against each other. Level-set queries are evaluated by transforming the points into the body frame (inverse translation and rotation), and, in periodic directions, by taking the minimum image of the point about the current centre, so that a body may cross a periodic boundary. The same transform gives $\mathbf{u}_b(\mathbf{x}) = \mathbf{U} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c}(t))$ at the cell centres. The kinematics may also be set from outside at every step, which is the hook for a body driven by its own equation of motion.

6 Passive scalar transport

Integrating (2) over the fluid part of a cell with the same arguments as above gives

$$\begin{aligned} \frac{(\alpha_P^{n+1} T_P^{n+1} - \alpha_P^n T_P^n) V_P}{\Delta t} + \sum_f \phi_f \bar{T}_f - \sum_f \theta_f D_f \delta_f |\mathbf{S}_f| (T_N - T_P) \\ + D \frac{A_{\text{wall}}}{d_{\text{wall}}} (T_P - T_w) [\text{fixed value}] = q_P \alpha_P V_P, \end{aligned} \quad (11)$$

with α^n the swept fraction (9) for a moving body. The wall segment is seen either as a fixed-value surface — the analogue of no slip, one diagonal term with the Dirichlet wall coefficient of Section 3 and the source $DA_{\text{wall}}T_w/d_{\text{wall}}$ — or as a zero-flux surface, the analogue of free slip, which needs no term at all because the wall segment simply carries no flux. The two masks, for velocity and scalar, are independent. Solid cells are blanked to T_w so that a cell released by a moving body starts from a defined value. No mass-source term appears: Reynolds transport gives exactly (11) because the relative flux through the wall is zero, and a uniform T is preserved for the same reason as in Proposition 4. The source q multiplies α_P , so it never injects into the solid.

Cells containing the wall are fluid cells: the scalar is transported into their fluid part through the wet fraction of their faces and never across the wall segment. The budget

$$\begin{aligned} \underbrace{\sum_P \alpha_P V_P T_P}_{\text{content}} - \underbrace{\sum_n \Delta t \sum_P q_P \alpha_P V_P}_{\text{injected}} + \underbrace{\sum_n \Delta t \sum_P D \frac{A_{\text{wall}}}{d_{\text{wall}}} (T_P - T_w)}_{\text{through walls}} \\ + \underbrace{\sum_n \sum_P (\alpha_P^{\text{prev}} - \alpha_P^n) V_P T_P^n}_{\text{swept defect}} = \text{round-off} \end{aligned} \quad (12)$$

is evaluated at every step. The last term exists only for moving bodies: α^n is integrated from the wall flux and is first-order consistent with the geometric fraction α^{prev} of the previous step,

and a cell absorbed into the body hands over no content. It is the price of the exact geometric conservation law of Section 5.2, and it is small (Section 8.3).

7 Implementation in OpenFOAM

7.1 Explicit operators on an unmodified mesh

The mesh is left untouched; θ_f and α_P enter the standard operators explicitly. The momentum equation of the transient solver reads

```
fvm::ddt(ibm.alpha(), U) // alpha V/dt; alpha.oldTime() = alpha^n
+ fvm::div(phi, U) // phi = theta (U_f . Sf) = ibm.flux(U)
- fvm::laplacian(ibm.thetaInterpolate(nu), U) // theta-weighted, no wall term
+ fvm::Sp(nu*ibm.noSlipCoeff(), U) // wall traction, no-slip bodies
+ fvm::Sp(nu*ibm.blankCoeff(), U) // blanking of solid cells
== fvOptions(U) - ibm.grad(p) // sum_f theta_f (p_f - p_P) Sf / V
```

and, for moving bodies, `UEqn -= ibm.movingWallSource(nu)` adds $\nu(\text{noSlipCoeff} + \text{blankCoeff})\mathbf{u}_b$. The pressure equation is

```
fvm::laplacian(ibm.thetaInterpolate(rAtU), p) == fvc::div(phiHbyA) // phiHbyA = ibm.flux(HbyA)
pEqn -= ibm.massSource(); // moving bodies only: div(phi) = -S
ibm.constrainPressure(pEqn); // regular rows for the solid cells
```

and `pEqn.flux()` is then automatically θ -weighted. Everything else is untouched `simpleFoam` and `pimpleFoam`: the boundary constraints on $\mathbf{H}/a_P$ and p , the pressure reference, pressure relaxation, `fvOptions`, residual control and the PIMPLE loops.

This choice keeps the full cell volume V_P in `fvMatrix::A()`, so that $r_A = V_P/a_P$ is exactly the mobility of (6). Scaling the face area vectors and cell volumes of the mesh in place would make all operators θ/α -aware implicitly, but would change the SIMPLEC mobility to $\alpha V_P/a_P$, break the interpolation weights on closed faces and require floors for solid cells; explicit operators are transparent and term-for-term identical to the formulation above.

The two-level unsteady term of Table 2 costs no solver code: the density-weighted time derivative of OpenFOAM uses the old-time value of the weight in its source, so the swept fraction α^n of (9) is simply stored as the old-time value of α after the geometry update. The transient Rhie–Chow correction (`ddtCorr`) is provided in Euler form with the old flux rescaled to the new wet area.

7.2 Structure

ibmBody a body: analytical (cylinder, sphere, box, half space), any closed `searchableSurface` (e.g. an STL through `triSurfaceMesh`), or a supplied fraction field; with its wall conditions and its `bodyMotion` (none, translating, oscillating, prescribed `Function1`, external).

cutCellGeometry the fractions and derived fields of Section 3, the cut-cell operators (gradient, flux, `ddtCorr`), the solid-row regularisation, the forces, and `update()`, which returns immediately unless a body moves. The moving-only fields (body velocity, mass source) are allocated only then, so the static path carries no additional cost.

ibmMeanVelocityForce constant-flow-rate forcing for periodic domains: the bulk velocity is averaged over αV , the force acts on the fluid volume only, and the controller gain is the SIMPLEC mobility $1/(A - H_1)$, the exact bulk response to a uniform force, floored at $\alpha/\Delta t$; $1/A$ underestimates the response by the diffusion number $\nu\Delta t/h^2$ with implicit viscosity. A zero target is allowed.

ibmScalarTransport equation (11), the per-body wall masks, a circular source, and the budget (12).

`ibmSimpleFoam`, `ibmPimpleFoam` `simpleFoam` and `pimpleFoam` with the terms above; the moving-body operations sit behind three `if (ibm.moving())` guards in the transient solver.

The geometry computation is local except for the synchronisation of coupled faces (`syncTools`) and global reductions, so parallel runs need nothing else; a four-rank run reproduces the serial forces to 2×10^{-12} .

8 Verification

All cases use a circular cylinder of radius $R = 1$ in a 10×10 doubly periodic box discretised by 81×81 cells, $\nu = 1$, $\alpha_{\min} = 0.02$, central differencing for convection. The reference is an independent implementation of the same discretisation on a structured grid (direct sparse solves, exact pinning of the bulk velocity by superposition), so that the comparison isolates the OpenFOAM implementation from the formulation.

8.1 Body at rest

The flow is driven at bulk velocity 0.5 ($Re_D = 1$). Table 3 compares geometry, fields and drag. The fractions agree to round-off (α exactly, θ to 10^{-11} , the bisection tolerance). The transient solver run with the reference’s pseudo-time step ($\Delta t = 65h^2/\nu$, Euler, one corrector, SIMPLEC, no `ddtCorr`) converges to the same discrete fixed point (velocity to 10^{-5} , drag to 2×10^{-6}); the steady solver differs through the relaxation dependence of the Rhie–Chow dissipation. Bodies given as an STL (a 3600-gon) or as a fraction field give the expected accuracy of their geometry route. With two bodies of mixed wall conditions the forces sum to the driving force times the fluid volume to all digits.

Table 3: Body at rest, no slip unless stated; reference drag 5.660008 (no slip), 2.935529 (free slip), per unit depth and density.

	steady	transient, ref. Δt	STL body	fraction field	free slip
α , cell by cell	exact	exact	exact	exact	exact
θ , face by face, max diff.	7.6×10^{-11}	7.6×10^{-11}	6.4×10^{-6}	0.11	7.6×10^{-11}
u , max diff. / max u	2.2×10^{-4}	9.4×10^{-6}	2.2×10^{-4}	3.3×10^{-3}	1.7×10^{-5}
drag	5.660383	5.659997	5.660383	5.651222	2.935496

8.2 Moving body: free stream, oscillation, Galilean invariance

Free-stream preservation. The domain is filled with $\mathbf{u} = \mathbf{u}_b$ and the body translates at $\mathbf{u}_b$ for five steps of $\Delta t = 0.025$ (Proposition 4). The maximum departure from $\mathbf{u}_b$ over the fluid is 0.0 in u and below 2×10^{-15} in v for $u_b = 0.37$ and $u_b = 2.0$ and for one or three outer iterations; the reference gives 6×10^{-16} and 4×10^{-15} . The test exercises the two-level unsteady term, the mass source, the moving-wall traction, the blanking and the flux re-interpolation simultaneously; a sign error in any of them gives an $O(1)$ result.

Oscillating cylinder. $x_c = 5 + \sin(2\pi \cdot 0.05 t)$, $t \in [0, 10]$, 400 steps, three outer iterations, bulk velocity pinned to zero. The force histories agree to 4.4×10^{-4} after the impulsive first ten steps (7×10^{-5} of the peak force; the first step, $F = -60.6$, agrees to 0.2%), the driving force to 10^{-6} , and the final fields to 2×10^{-7} in velocity (Fig. 4). The mass-source sum and the continuity error $\max_P |\sum_f \phi_f + S_P|$ are at round-off at every step.

Galilean invariance. A body translating at $u_b = 0.5$ through a fluid of zero bulk velocity is the Galilean image of the body at rest in a bulk flow of 0.5, but is computed through entirely different terms. Over $t \in [0, 30]$ the body crosses the periodic seam twice. The mean force over the last 400 steps is -5.69099 ± 0.0883 , against -5.69103 ± 0.0884 for the reference and $+5.66038$ for

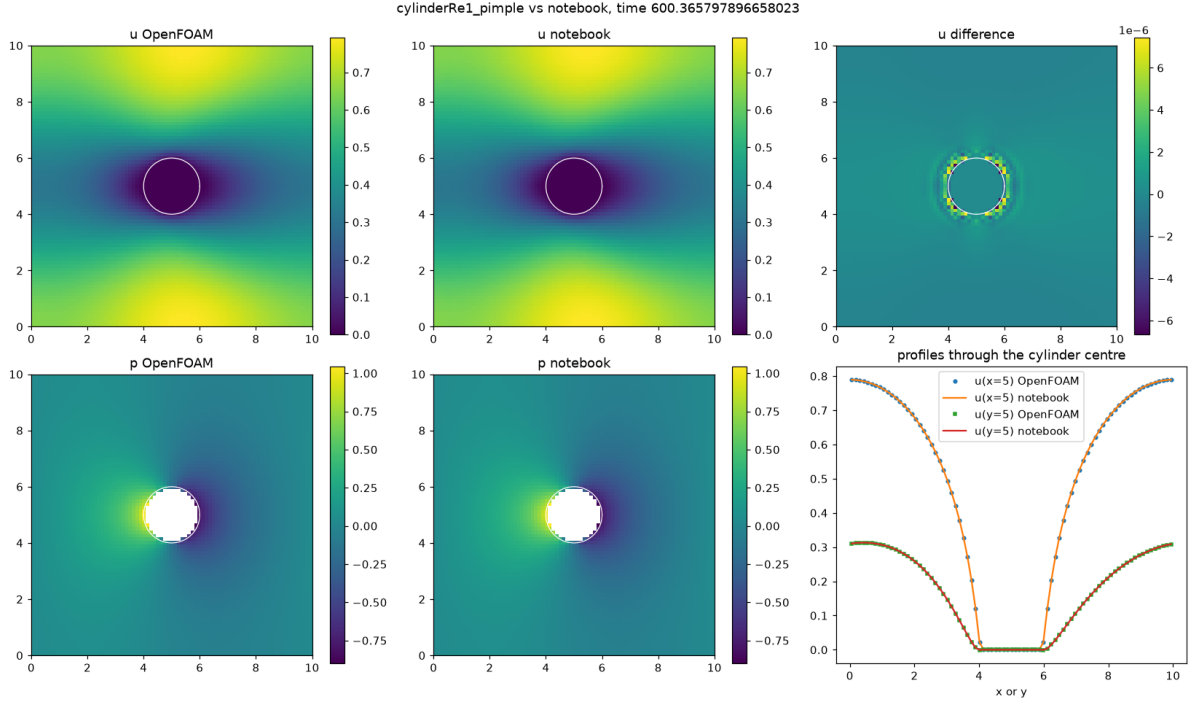


Figure 3: Body at rest, transient solver with the reference's pseudo-time step: streamwise velocity (top) and pressure (bottom) against the reference, difference field, and profiles through the cylinder centre.

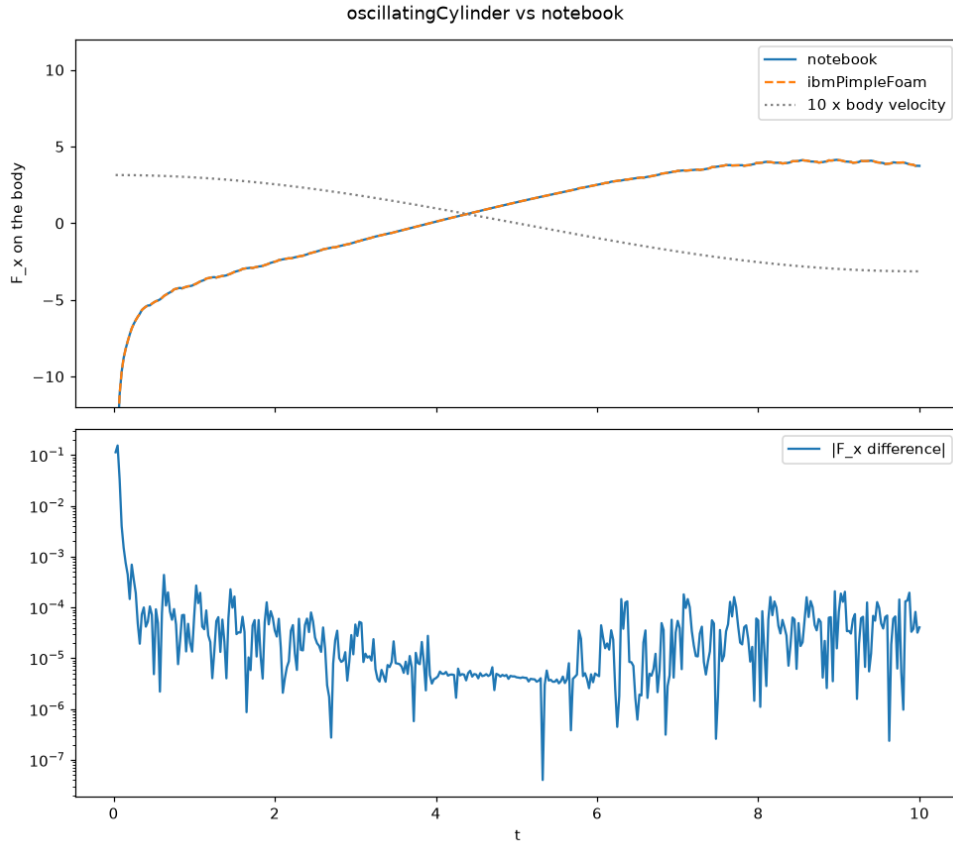


Figure 4: Oscillating cylinder: streamwise force on the body against the reference (top) and their difference (bottom).

the body at rest: the 0.5% difference between moving and fixed body is the first-order signature of the $O(\Delta t)$ swept-volume integration, and the $\pm 1.5\%$ ripple is the body sliding from one cut pattern to the next, the characteristic signature of a cut-cell method on a non body-fitted mesh. The histories agree to 1.3×10^{-3} at every step after the impulsive start (Fig. 5).

This test exposed a subtlety of the seam treatment. The reference, as published, copies θ across the periodic seam *before* absorbing slivers, so an absorbed cell at $x = L$ closes its face on that side while the same face stays open for the cell at $x = 0$; while the body front or back crosses the seam its force deviates by up to 2.7% for a few steps. Synchronising the coupled faces with the minimum *after* absorption (Section 3.2) keeps closure exact, the mass-source sum being 10^{-19} throughout; applying the same rule to the reference removes the deviations, and the numbers above are against that reference.

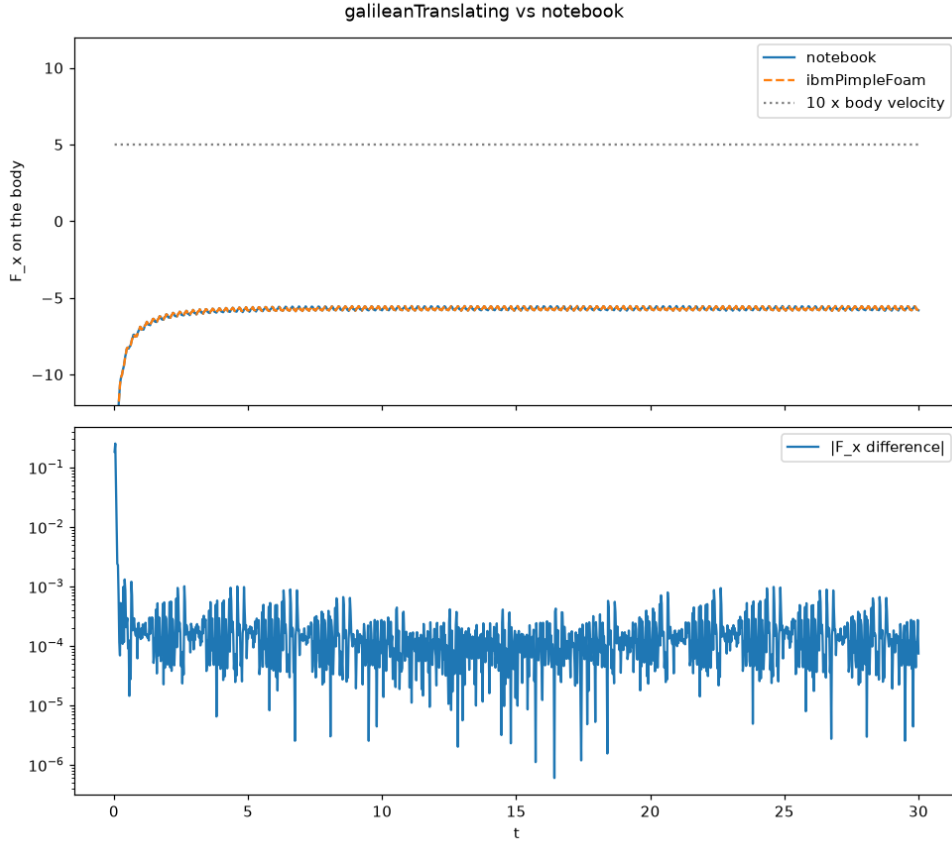


Figure 5: Translating cylinder crossing the periodic seam twice: force history against the reference (top) and their difference (bottom).

8.3 Passive scalar

A scalar initially zero is injected at unit rate in a disk of radius 0.5 centred at (5, 8.5), away from the path of the body, with $D = 0.1$, and the budget (12) is followed (Table 4, Fig. 6). With zero-flux walls and a body at rest the content equals the injected amount to 4×10^{-12} after 200 steps and the solid stays exactly at zero: nothing is lost into the solid. With fixed-value walls the amount that left through the wall closes the budget to 6×10^{-11} , and in the steady solver the source rate equals the wall flux rate to 10^{-11} . With a moving body the content differs from the injected amount by the swept-volume defect only, 3×10^{-5} of the content for the oscillating cylinder and 8×10^{-5} for the translating one, the solid stays at the wall value, and the residual budget errors of 10^{-10} are accumulated linear-solver tolerance. The flow is unaffected by the scalar.

Table 4: Scalar budget (12) at the end of each run.

case	steps	content	injected	through walls	swept defect	budget error
at rest, zero flux (transient)	202	19.581366	19.581366	0	0	-3.9×10^{-12}
at rest, fixed value	202	11.904949	19.581366	7.676417	0	-5.7×10^{-11}
at rest, fixed value (steady)	1834	16.323010	rate 0.097847	rate 0.097847	—	-1.0×10^{-11}
oscillating, zero flux	400	0.978439	0.978472	0	3.3×10^{-5}	-8.0×10^{-11}
oscillating, fixed value	400	0.968143	0.978472	0.010344	-1.5×10^{-5}	-6.1×10^{-11}
translating, zero flux	1200	2.935650	2.935415	0	-2.3×10^{-4}	-7.5×10^{-10}

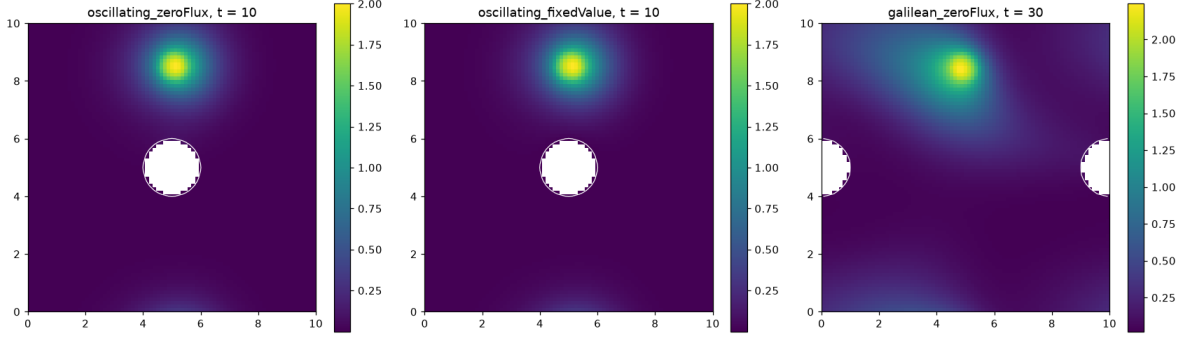


Figure 6: Scalar field for the oscillating cylinder with zero-flux (left) and fixed-value (centre) wall, and for the translating cylinder with zero-flux wall (right); the body is blanked.

9 Second-order wall treatment

The no-slip treatment of Section 4 is first order. An order study on manufactured solutions (Poisson and convection–diffusion problems on the interior of a disk with Dirichlet and Neumann walls, `doc/secondOrderStudy.md`) shows that the limiting error is not the two-point wall gradient but the diffusive flux across the faces of a cut cell, $\theta_f \Gamma_f(u_N - u_P)/h$, which assumes the two cell values to sit h apart along the face normal whereas they represent fluid parts whose centroids are neither. Two ingredients restore second order: (i) the wall distance d_c of the fluid centroid to the wall instead of (4), and (ii) a face flux consistent with the centroids,

$$\mathbf{n} \cdot \nabla u|_f = \frac{u_N - u_P}{\mathbf{n} \cdot \mathbf{d}} + \left[\mathbf{g}_f \cdot \mathbf{n} - \frac{\mathbf{g}_f \cdot \mathbf{d}}{\mathbf{n} \cdot \mathbf{d}} \right], \quad (13)$$

with $\mathbf{d}$ the centroid-to-centroid vector and $\mathbf{g}_f$ the interpolated least-squares gradient on the centroids; the first term is implicit, the bracket explicit (the over-relaxed non-orthogonal form, exact for a linear field and zero on uncut cells). No probe into the fluid, no ghost value and no stencil growth are needed, and the free-slip wall is already second order. In OpenFOAM, (13) is one `snGradScheme` (`cutCellCorrected`) supplying the delta coefficients $1/(\mathbf{n} \cdot \mathbf{d})$ and the explicit remainder, which the Gauss Laplacian, its flux and `fvc::snGrad` then use consistently; the cut-cell gradient (5) evaluates p_f at the wet-face centroid and p_{wall} by extrapolation from the centroid. The treatment is a run-time switch and leaves the first-order path bit-identical.

On the manufactured problems ($N = 20 \dots 160$) the L_2 orders become 1.92, 2.06, 2.09 (Dirichlet Laplacian) and 1.76, 1.94, 1.99 (Dirichlet convection–diffusion) against 0.65, 1.22, 1.04 and 0.55, 1.04, 1.21 before, with errors two orders of magnitude smaller at $N = 160$; the Neumann problems stay at 1.8–2.5. In the coupled problem, Taylor–Couette flow between two immersed cylinders (inner rotating, no slip; $N = 40, 80, 160$), the velocity error drops by 5–16 times and converges at order 1.7 in the bulk (1.8 then 1.4 overall), while the pressure of the cut cells does not converge: the mobility of a cut cell scales with $\alpha_P \theta_f$ because the wall term dominates its diagonal, so its pressure is set by tiny flux residuals — the small-cell problem of the pressure. It pollutes the near-wall pressure and the forces at first order: the cylinder drag of Section 8 converges with order 1.7 (no slip) and 0.8 (free slip) by Richardson extrapolation on $N = 41, 81, 161$, against 0.3

and 0.8 before, and the no-slip sequence does not keep its rate on a fourth mesh. Second-order forces need a cut-cell pressure that is not determined by the cut cell alone (cell merging, or a wall pressure formed from the full cells), which is the natural next step. The same test exposed and fixed a defect of the moving-body path: the wall velocity of a rotating body must be evaluated at the wall segment, not at the cell centre, or $\mathbf{u}_b \cdot \mathbf{S}_w$ does not vanish for a tangential motion.

10 Limitations

- *Order.* First order in time (implicit Euler, inherited by the swept-volume integration); first order in space for no slip with the treatment of Section 4, second order for the operators and the bulk velocity with that of Section 9, forces first order in both. A second-order extension in time would need α at three levels and a three-level form of (9).
- *Slivers.* Absorption below $\alpha_{\min}$ displaces the boundary by $O(\alpha_{\min}h)$, and a cell crossing the threshold hands over no momentum or scalar content; mass conservation is unaffected. Cell merging is the clean remedy and would also relax the step restriction.
- *Laminar.* Turbulence models are constructed for their effective viscosity only; their own transport equations would need the same treatment as the scalar.
- *Fraction-field bodies.* Reconstructing θ from the 0.5 iso-surface of the point-interpolated fraction smears the wall over one cell (drag -0.16% on the cylinder); a per-cell iso-value search reproducing each cell's α would bring it to $O(h^2)$.
- *One fluid region.* A single pressure reference; a body splitting the fluid would need one per region.
- *Cost of moving bodies.* The level set is evaluated at all points, face and cell centres every step, and n_s^d samples per cut cell; for three-dimensional triangulated bodies a smaller n_s or a band around the previous position is advisable.

11 Conclusions

The method rests on one geometric statement, the closure identity of a cut cell, and on one choice for moving bodies, the wall flux as the primary quantity. From the first follow exactness for uniform tangential flow, the absence of spurious pressure forces and no-penetration itself; from the second follow exact compatibility of the pressure system and a discrete geometric conservation law that preserves a free stream to machine precision. The whole boundary treatment adds to the interior scheme two scalar fields, θ_f and α_P , one diagonal coefficient for a no-slip or fixed-value wall, and one source term for a moving wall. Implemented in OpenFOAM with explicit operators on an unmodified mesh, it reproduces an independent reference implementation to the level of the linear-solver tolerances for a body at rest, oscillating and translating through a periodic boundary, and transports a passive scalar with a budget that closes to round-off.